Syllogistic Entailment for Distributional Models

Caleb Kisby
M781/L715

February 20, 2022

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-

Distributional Semantics

... of anticipating being hurt by another	horse	especially if some other rider comes ...
... was simply beaten by a better	horse	at the distance on the day ...
... can infer from these studies that	horses	reared with other horses in a ...
... studies that horses reared with other	horses	in a free and enriched environment ...
... people saying "Is that all your	horse	gets to eat?" in amazement. The ...
... and a cache of cattle and	horse	bones, while from Normangate Field a ...
... Bachelor's Button was a sterling good	horse,	especially at Ascot, but he was ...
... the same way as a domestic	horse	that it may be stabled with ...
... Attachment in 1790 – that is, one	horse	or two cows for each £4 ...
... grey hair as coarse as a	horse	's tail straying from her mob-cap ...

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$$\vec{v}_{\text{horse}} = \langle \underset{\text{Button}}{9} \quad \underset{\text{reared}}{4} \quad \underset{\text{rider}}{33} \quad \underset{\text{especially}}{1} \quad \ldots \rangle$$

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  BIG OPEN PROBLEM

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```
woman
['republic', 'michelle', 'sarah_palin', 'westerner', 'tornado',
'kingfisher', 'airbus', 'tonga', 'democrat', 'michael_jackson',
'sgt', 'clinton', 'grand_slam', 'hiv', 'lindsay_lohan', 'chihua
'parasol']
```

Testing (and Lack Thereof)

- Ran out of time to do big, systematic tests
- But did this small test by hand (wrote out all the inclusions between these words)
- 86% acc for determining non-subsets
- 40% acc for determining subsets
- To illustrate what kind of problems we run into, I thought you all might like to see:
- All boys are children, as we expect
- and All girls are children, that's nice
- but all girls are boys! Definitely not true
- What went wrong? In our corpus, it is likely that most contexts that involve 'girl' also include 'boy', but then 'boy's are talked about in many places where 'girl' is not

- Obviously, this isn't sufficient for a conference. So I have planned two kinds of big-scale tests:
- The first is to generate instances of rules for All and other relations, and then check the accuracy of our system in determining whether these rules hold
- The second is to check our system's "world knowledge" by doing a more systematic version of our by-hand test

Small, by-hand test:

```
categories = ["president", "politician",
    "boy", "girl", "child",
    "fish", "bird", "animal", "species",
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Key References

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